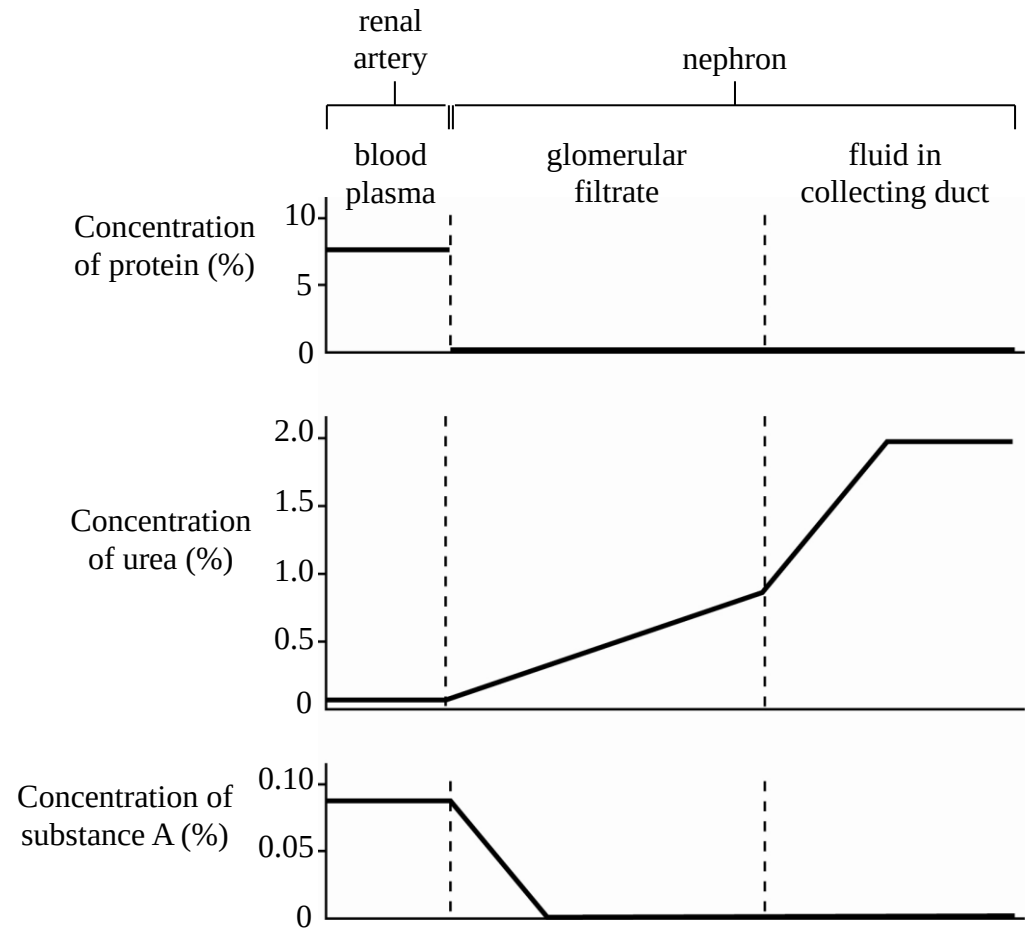
## Structured questions

|!|BCELB0330000001|!| ★

The graph below shows the changes in the concentration of protein, urea and substance A as they flow from the renal artery to the nephron.




(a) Describe and explain the difference in the concentration of protein between the blood plasma and the glomerular filtrate. (2 marks)

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(b) What process causes the increase in the concentration of urea along the collecting duct? (1 mark)

(c) (i) Suggest what substance A could be. (1 mark)

(ii) Describe and explain the change in concentration of substance A along the nephron. (3 marks)

(d) If a person drinks a large amount of water, how would the concentration of urine be affected? Describe the mechanism that would bring about this result. (6 marks)

##

(a) Proteins are present in the blood plasma within the renal artery but not in the glomerular filtrate. (1)
Proteins are too large to pass through the walls of the glomerulus and Bowman's capsule. (1)

(b) The reabsorption of water along the collecting duct (1)

(c) (i) Glucose / amino acid (1)

(ii) The concentration of substance A falls gradually to zero along the nephron. (1)
All of substance A has been reabsorbed into the bloodstream (1)
by diffusion and active transport. (1)

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(d) The concentration of urine would decrease. (1)
The water potential of blood increases after the intake of a large amount of water. (1)
The osmoreceptors in the hypothalamus detect the increase in the water potential of blood. (1)
The hypothalamus signals the pituitary gland to release less ADH. (1)
The wall of collecting ducts becomes less permeable to water. (1)
As a result, a smaller proportion of water is reabsorbed (1)
and a large volume of dilute urine is produced.

(13 marks)

##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Analysing data | 2 |
| (b) | Applying concepts | 1 |
| (c)(i) | Analysing data | 1 |
| (c)(ii) | Applying concepts | 3 |
| (d) | Applying concepts | 6 |

|!|BCELB0330000002|!| ★

The table below lists Cindy's daily water gain and water loss.

| Water gain | Volume ($cm^3$) | Water loss | Volume ($cm^3$) |
|---|---|---|---|
| Drink | 1600 | Urine | 1450 |
| Food | 900 | Faeces | 100 |
| Water released in chemical reactions | 350 | Exhaled air | 450 |
| | | Sweat | ? |
| Total | 2850 | Total | 2850 |

(a) Using the table above, calculate the volume of water loss due to sweating. (1 mark)

(b) Name the organ that performs the following function:
(i) exhalation (1 mark)

(ii) urine formation (1 mark)

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(iii) faeces formation (1 mark)

(iv) sweating (1 mark)

(c) On a hot day, Cindy sweated a lot.

(i) Describe how sweating affects the water potential of her blood. (2 marks)

(ii) State and explain how the change in (c)(i) affects the volume of urine formed. (5 marks)

##

(a) 850 $cm^3$ (1)

(b) (i) lung (1)

(ii) kidney (1)

(iii) colon (1)

(iv) skin (1)

(c) (i) Sweating increases water loss of the body. (1)

The water potential of blood falls. (1)

(ii) The osmoreceptors in the hypothalamus detect the fall in the water potential of the blood. (1)

The hypothalamus stimulates the pituitary gland to release more ADH into the blood. (1)

More ADH causes the wall of the collecting ducts to become more permeable to water. (1)

A larger proportion of water is reabsorbed into the blood. (1)

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A smaller volume of concentrated urine is formed. (1)

(12 marks)

##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Mathematical skills | 1 |
| (b)(i) | Understanding basic concepts | 1 |
| (b)(ii) | Understanding basic concepts | 1 |
| (b)(iii) | Understanding basic concepts | 1 |
| (b)(iv) | Understanding basic concepts | 1 |
| (c)(i) | Applying concepts | 2 |
| (c)(ii) | Applying concepts | 5 |

|!|BCELB0330000003|!| ★ **Cross topic**

The table below lists Cindy's daily water gain and water loss.

| Water gain | Volume ($cm^3$) | Water loss | Volume ($cm^3$) |
|---|---|---|---|
| Drink | 1600 | Urine | 1450 |
| Food | 900 | Faeces | 100 |
| Water released in chemical reactions | 350 | Exhaled air | 450 |
| | | Sweat | 850 |
| Total | 2850 | Total | 2850 |

(a) Describe how water is lost through exhalation. (2 marks)

(b) Cindy went jogging yesterday.

(i) She noticed that she had sweated much more than usual. What are the cause and effect of increased sweating during jogging? (4 marks)

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(ii) After jogging, Cindy felt thirsty and drank 1 L of water. Explain how this affected her urine output. (6 marks)

##

(a) Water evaporates from the moist respiratory tract and the air sacs of the lungs. (1)
The water vapour is breathed out in exhaled air. (1)

(b) (i) During jogging, the increased muscle contraction produces more heat. (1)
The body temperature increases when the rate of heat gain is greater than the rate of heat loss. (1)
The body produces more sweat to be evaporated from the skin surface so as to remove more heat. (1)
This helps to prevent the body from overheating. (1)

(ii) The water potential of her blood increased after she drank 1 L of water. (1)
The osmoreceptors in the hypothalamus detect the increase in the water potential of the blood. (1)
The hypothalamus signals the pituitary gland to release less ADH into the blood. (1)
The wall of collecting ducts becomes less permeable to water. (1)
As a result, a smaller proportion of water is reabsorbed (1)
and a large volume of dilute urine is produced. (1)

(12 marks)

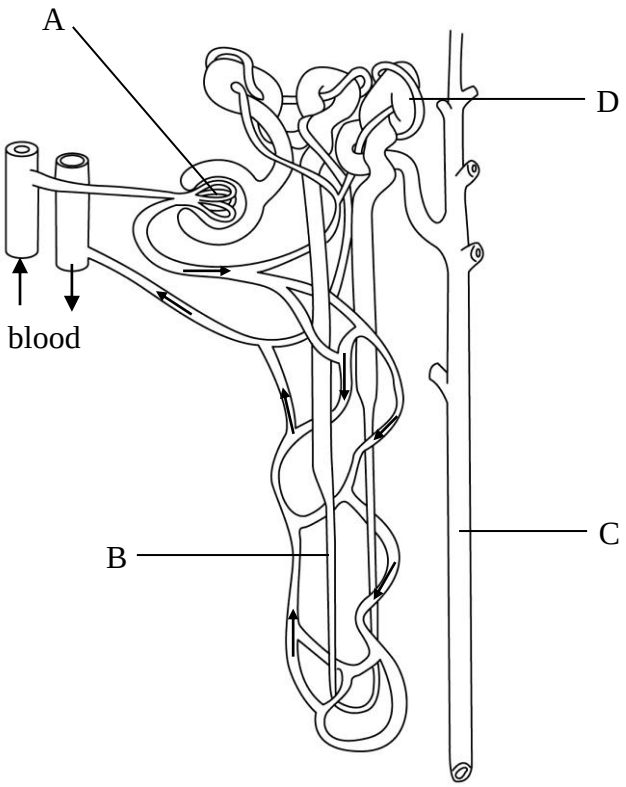
##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Understanding basic concepts | 2 |
| (b)(i) | Applying concepts | 4 |
| (b)(ii) | Applying concepts | 6 |

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|!|BCELB0330000004|!| ★

The diagram below shows a nephron. The daily composition of the glomerular filtrate and urine is shown in the following table.




| Substance | Glomerular filtrate | Urine |
|---|---|---|
| Water | 160 L | 1.5 L |
| Salts | 1000 g | 20 g |
| Glucose | 180 g | 0 g |
| Urea | 38 g | 21 g |

(a) Referring to the diagram, identify the structure involved in the formation of glomerular filtrate using the appropriate letter label. (1 mark)

(b) (i) Calculate the mass of salts that is reabsorbed each day from the nephron. (1 mark)

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(ii) Why is the reabsorption of salts important? (1 mark)

(c) (i) Calculate the concentration of urea (to 2 decimal places) in the glomerular filtrate and urine, respectively. (2 marks)

(ii) State and explain the difference between the two concentrations in (c)(i). (3 marks)

(d) State and explain in general the difference in urine concentration between a desert animal and a polar animal in their natural habitats. Assume that both take in an equal amount of water. (3 marks)

##

(a) Structure A (1)

(b) (i) (1000 – 20) g = 980 g (1)

(ii) Salts are essential for maintaining the water potential of the tissue fluid that bathes cells. (1)

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(c) (i) Concentration of urea in the glomerular filtrate = 38 g / 160 L = 0.24 g / L (1)

Concentration of urea in urine = 21 g / 1.5 L = 14 g / L (1)

(ii) Urine contains a higher concentration of urea (1)

because a greater proportion of water in the glomerular filtrate would be reabsorbed along the kidney tubule, (1)

while about 50% of the urea would not be reabsorbed and flows along the collecting duct. (1)

(d) The desert animal excretes urine that is more concentrated than that of the polar animal. (1)

Living in a hot and dry environment, much of the water taken in by the desert animal would be lost through sweating. (1)

The desert animal conserves water by reducing water loss through urine. (1)

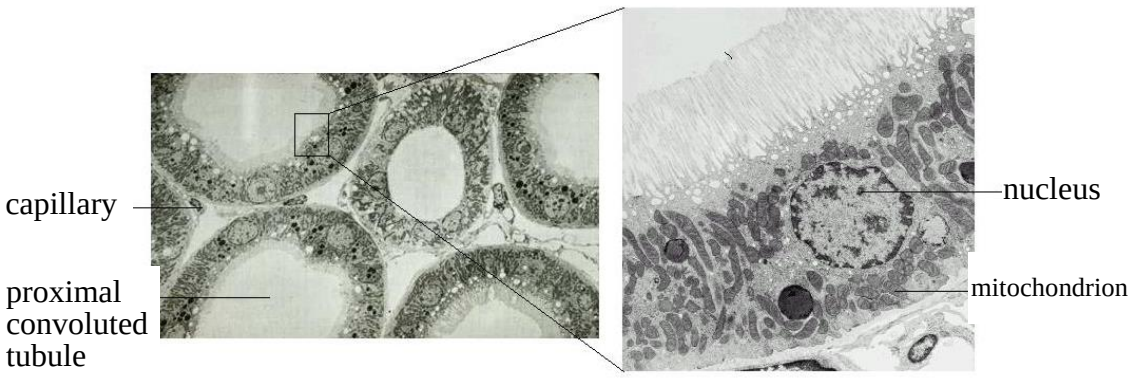
(11 marks)

##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Analysing data | 1 |
| (b)(i) | Mathematical skills | 1 |
| (b)(ii) | Understanding basic concepts | 1 |
| (c)(i) | Mathematical skills | 2 |
| (c)(ii) | Applying concepts | 3 |
| (d) | Applying concepts | 3 |

|!|BCELB0330000005|!| ★

The following micrographs show the cross section of a proximal convoluted tubule and an adjacent capillary.




Epithelial cells of the wall of the proximal convoluted tubule

(a) Name the fluid found in the lumen of the proximal convoluted tubule. (1 mark)

(b) List ***two*** organic compounds that can be reabsorbed into the capillary. (2 marks)

(c) State ***two*** transport mechanisms involved in the reabsorption of the two compounds you have given in (b). (2 marks)

(d) With reference to the micrographs, state and explain ***two*** structural features of the epithelial cells that help the reabsorption of the compounds you have given in (b). (4 marks)

##

(a) Glomerular filtrate (1)

(b) Amino acids (1)

Glucose (1)

(c) Diffusion (1)

Active transport (1)

(d) Feature 1: presence of numerous microvilli on the lumen-facing side of the cells (1)

Explanation: microvilli increase the surface area available for the reabsorption of the compounds (1)

Feature 2: presence of many mitochondria (1)

Explanation: the mitochondria provide a large amount of energy via respiration to drive the active transport of the compounds. (1)

(9 marks)

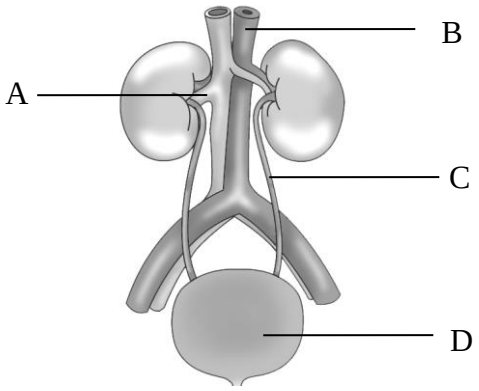
##

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| | Skills assessed | Marks |
|---|---|---|
| (a) | Understanding basic concepts | 1 |
| (b) | Understanding basic concepts | 2 |
| (c) | Understanding basic concepts | 2 |
| (d) | Analysing data | 2 |
| | Understanding basic concepts | 2 |

|!|BCELC0330000006|!|

The following diagram shows part of the human urinary system.




(a) Name vessels A and B. State and explain which one, A or B, contains a lower concentration of urea. (4 marks)

(b) Name the organic compounds from which urea is produced. State the organ responsible for urea formation. (2 marks)

(c) The fluid collected from vessel C does not contain glucose and proteins. Explain why and provide the underlying mechanism where possible. (2 marks)

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(d) The size of structure D can be changed to suit physiological needs. State one situation in which structure D expands and another in which it contracts. (2 marks)

##

(a) A: renal vein (1)
B: descending aorta (1)
Vessel A contains a lower concentration of urea (1)
because it receives blood from the kidney, where some urea is excreted. (1)
(b) Urea is produced from the breakdown of (excess) amino acids. (1)
The organ responsible for urea formation is the liver. (1)
(c) Glucose in the glomerular filtrate is reabsorbed into the blood by diffusion and active transport. (1)
Proteins in the blood are too large to pass through the walls of the glomerulus and the Bowman's capsule. (1)
(d) Structure D expands during the temporary storage of urine (1)
while it contracts during the discharge of urine / urination. (1)

(10 marks)

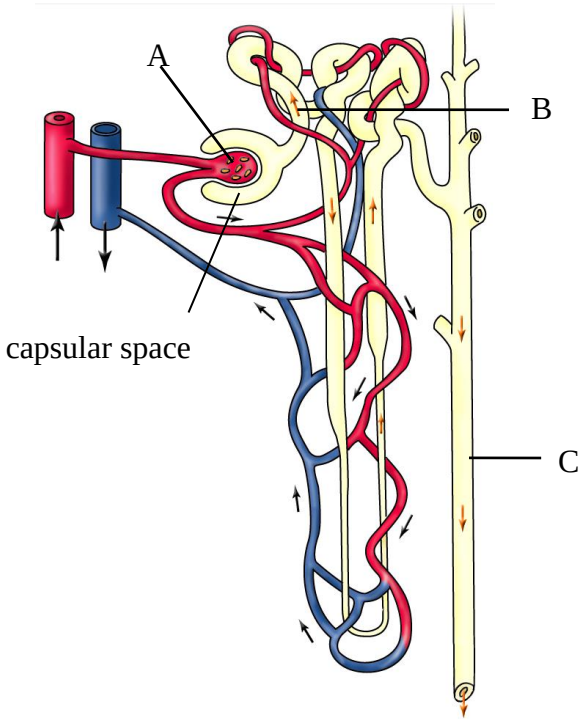
##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Analysing data | 2 |
| | Applying concepts | 2 |
| (b) | Understanding basic concepts | 2 |
| (c) | Applying concepts | 2 |
| (d) | Understanding basic concepts | 2 |

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|!|BCELB0330000007|!| ★ **Cross topic**

The diagram below shows the structure of a nephron.




(a) Ultrafiltration occurs in structure A.

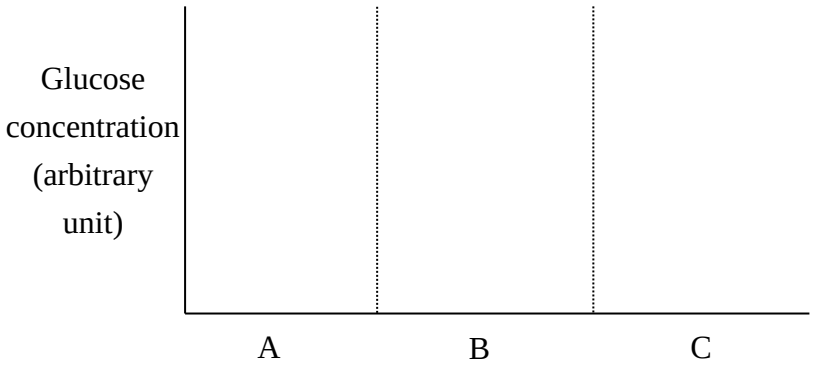
(i) Describe and explain this process. (3 marks)

(ii) Give ***two*** similarities between ultrafiltration and the formation of tissue fluid. (2 marks)

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(b) Structure B is composed of the proximal convoluted tubule, loop of Henle and the distal convoluted tubule.

(i) Sketch a graph to show the change in glucose concentration in the filtrate as it flows through structures A, B and C in a healthy person. (3 marks)




(ii) Explain the changes in your graph for (i). (3 marks)

(c) Explain how the concentration and the amount of urea in the filtrate changes as it flows from structure B to structure C. (3 marks)

##

(a) (i) Under the effect of high hydrostatic pressure in structure A, (1)
water and other small molecules (except plasma proteins) (1)
are forced out of the blood, exiting through the walls of structure A and the Bowman's capsule, and entering the capsular space. (1)

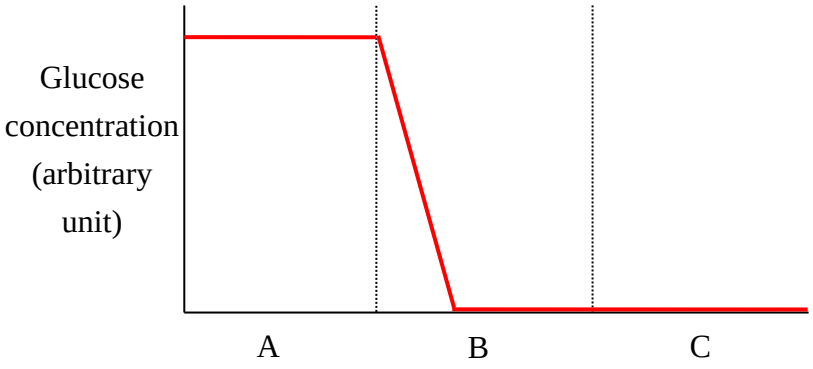
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(ii) Any ***two*** of the following: (2)

- High hydrostatic pressure forces out certain compounds from the capillaries.
- Capillary walls are used as a filter.
- Large plasma proteins / red blood cells / blood platelets are filtered out.

(Or other reasonable answers)

(b) (i)




Correct sketch for A (1)

Correct sketch for B (1)

Correct sketch for C (1)

(ii) Glucose concentration does not change between part A and the beginning of part B of the graph because glucose passes freely through the walls of structure A and the Bowman's capsule. (1)

Glucose concentration falls gradually to zero in part B of the graph as all the glucose is reabsorbed in structure B (1)

through diffusion and active transport. (1)

(c) The reabsorption of water between structures B and C will increase the urea concentration in the filtrate. (1)

Urea is partially reabsorbed (about 50%) due to its increased concentration. (1)

However, a greater proportion of water is reabsorbed, so urea concentration still increases in filtrate even as its overall amount decreases throughout structures B to C. (1)

(14 marks)

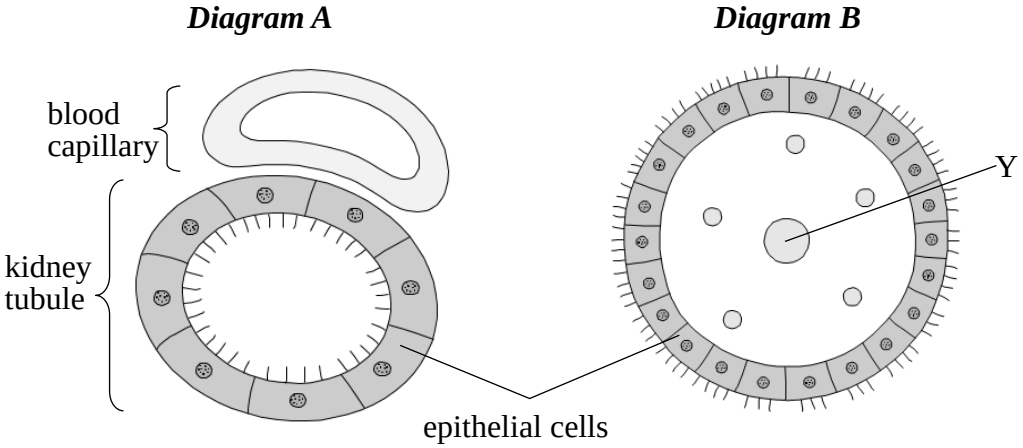
##

| | Skills assessed | Marks |
|---|---|---|
| (a)(i) | Understanding basic concepts | 3 |
| (a)(ii) | Applying concepts | 2 |
| (b)(i) | Presenting information | 3 |
| (b)(ii) | Applying concepts | 3 |
| (c) | Applying concepts | 3 |

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|!|BCELB0330000008|!| ★

The two diagrams below are ***not*** drawn to scale. **Diagram A** shows the transverse sections of a kidney tubule and an adjacent blood capillary. **Diagram B** shows a villus of the ileum.




(a) Name structure Y and the food substance it absorbs. (2 marks)

(b) With reference to the diagrams, state ***two*** structural similarities between the kidney tubule and the villus. (2 marks)

(c) Glucose enters the bloodstream through both reabsorption from the lumen of the kidney tubule and through absorption from the intestinal lumen.

(i) Besides diffusion, name another transport mechanism that facilitates glucose uptake in the reabsorption and absorption processes. (1 mark)

(ii) Explain how the reabsorption or absorption of glucose can bring about the passive movement of water molecules. (3 marks)

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(d) State ***one*** cellular adaptation in epithelial cells that facilitates the transport process you have given in (c)(i) during reabsorption and absorption. (1 mark)

##

(a) Structure Y: lacteal (1)
Food substance absorbed: lipids (1)

(b) Any ***two*** of the following: (2)
- They have extensive systems of capillaries either nearby or within.
- Their epithelial cells possess many microvilli / tiny outgrowths / projections.
- Their epithelia are one-cell-thick.

(c) (i) Active transport (1)

(ii) The transport of glucose raises the water potential in the lumen. (1)
This causes water to exit the lumen, move across the epithelial cells and into the blood capillaries (1)
by osmosis. (1)

(d) Epithelial cells have abundant mitochondria for driving active transport. (1)

(9 marks)

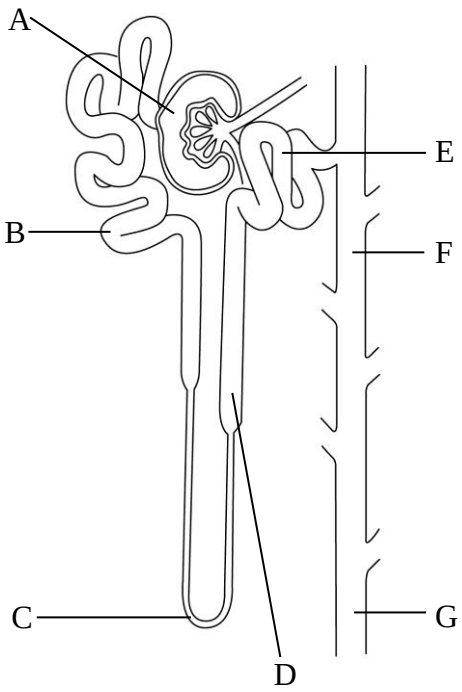
##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Analysing data | 1 |
| | Understanding basic concepts | 1 |
| (b) | Analysing data | 2 |
| (c)(i) | Understanding basic concepts | 1 |
| (c)(ii) | Understanding basic concepts | 3 |
| (d) | Applying concepts | 1 |

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|!|BCELB0330000009|!| ★

The diagram below shows a nephron.




(a) Name the regions labelled E and F. (2 marks)

_____

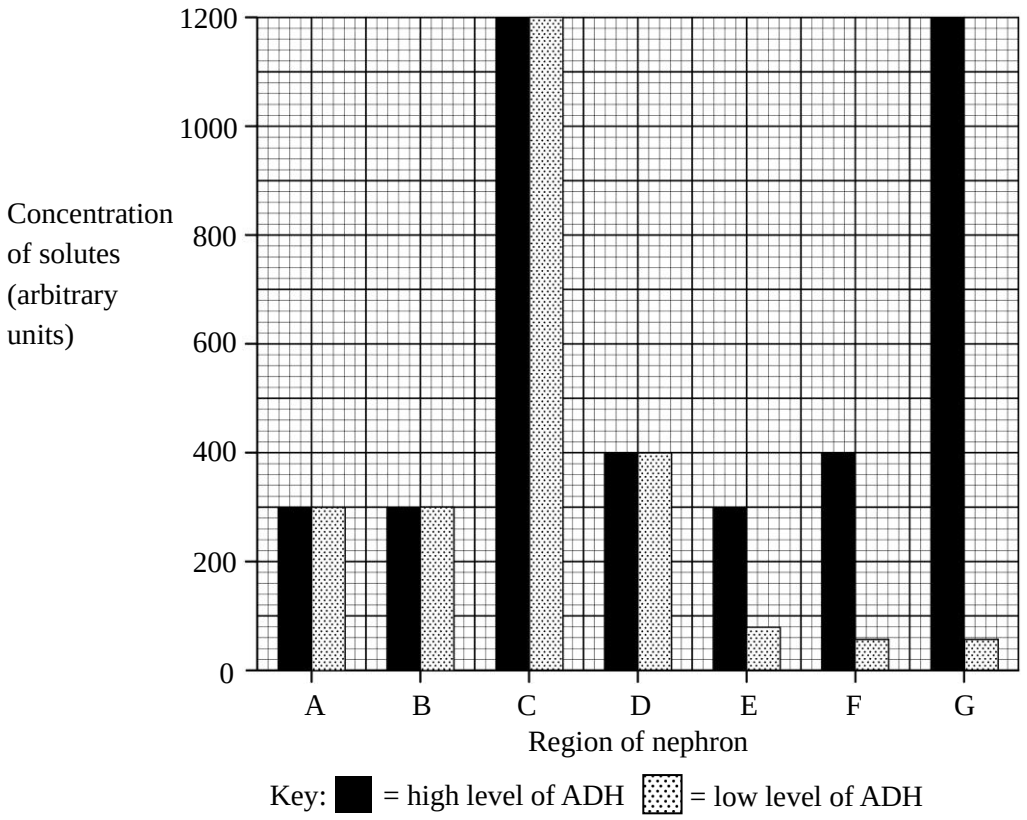
_____

(b) Where can nephrons be found in the kidney? (1 mark)

_____

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(c) The graph below shows the concentration of solutes in each of the labelled regions of the nephron when there is either a high level of antidiuretic hormone (ADH) or a low level of ADH in the blood.




(i) Calculate the percentage increase in the concentration of solutes between regions A and G when the level of ADH in the blood is high. Show your working. (2 marks)

(ii) Using the information in the graph and your knowledge of the hormonal control of osmoregulation, explain how a decrease in the water potential of the blood leads to the formation of a smaller volume of concentrated urine. (4 marks)

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(iii) Suggest ***one*** scenario that would cause the human body to form a smaller volume of concentrated urine. (1 mark)

##

(a) E: distal convoluted tubule (1)
F: collecting duct (1)

(b) The cortex and medulla (1)

(c) (i) (1200 – 300) / 300 × 100% (1)
= 300% (1)

(ii) Osmoreceptors in the hypothalamus detect the decrease in the water potential of the blood. (1)
The hypothalamus signals the pituitary gland to release more ADH into the blood. (1)
ADH causes the walls of the collecting ducts (structure G) to become more permeable to water. (1)
A larger proportion of water is reabsorbed into the blood. (1)
As a result, a smaller volume of concentrated urine is formed.

(iii) Heavy sweating / other reasonable answers (1)

(10 marks)

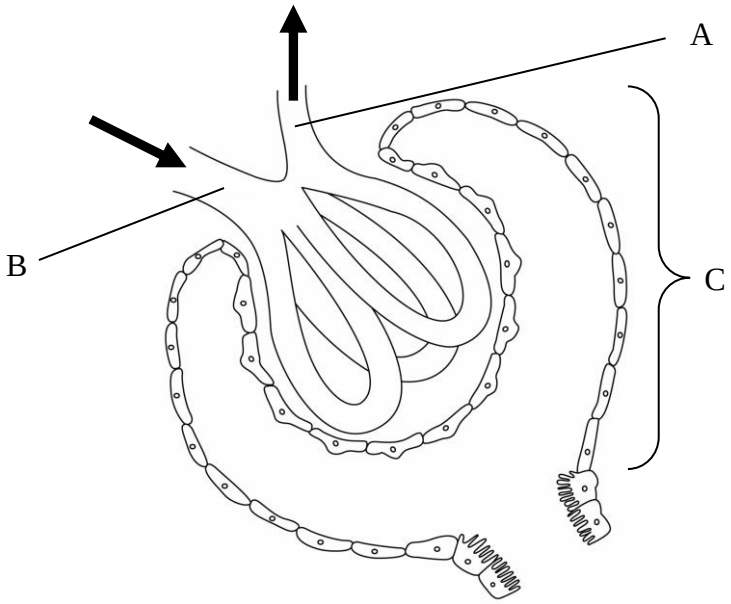
##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Analysing data | 2 |
| (b) | Understanding basic concepts | 1 |
| (c)(i) | Mathematical skills | 2 |
| (c)(ii) | Applying concepts | 4 |
| (c)(iii) | Applying concepts | 1 |

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|!|BCELC0330000010|!|

The following diagram shows the structure of a glomerulus, where ultrafiltration occurs in the kidney.




(a) Identify structures A and C. (2 marks)

(b) Besides water and glucose, suggest ***two*** other substances that can be found in the glomerular filtrate. (2 marks)

(c) Referring to the labelled structures, describe the process of ultrafiltration. (4 marks)

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(d) Explain how a large loss of blood from the body would affect the rate of glomerular filtrate formation. (2 marks)

_____

_____

##

(a) Structure A: efferent arteriole (1)
Structure C: Bowman's capsule (1)

(b) Any ***two*** of the following: (2)

- urea
- amino acids
- mineral salts

(Or other reasonable answers)

(c) The blood reaching the glomerulus from vessel B is under high hydrostatic pressure. (1)
As a result, water and other small molecules in the blood (1)
are forced out through the walls of the glomerulus and structure C (1)
to form a fluid called the glomerular filtrate in structure C. (1)

(d) The blood loss lowers the blood pressure, (1)
which in turn lowers the rate of glomerular filtrate formation. (1)

(10 marks)

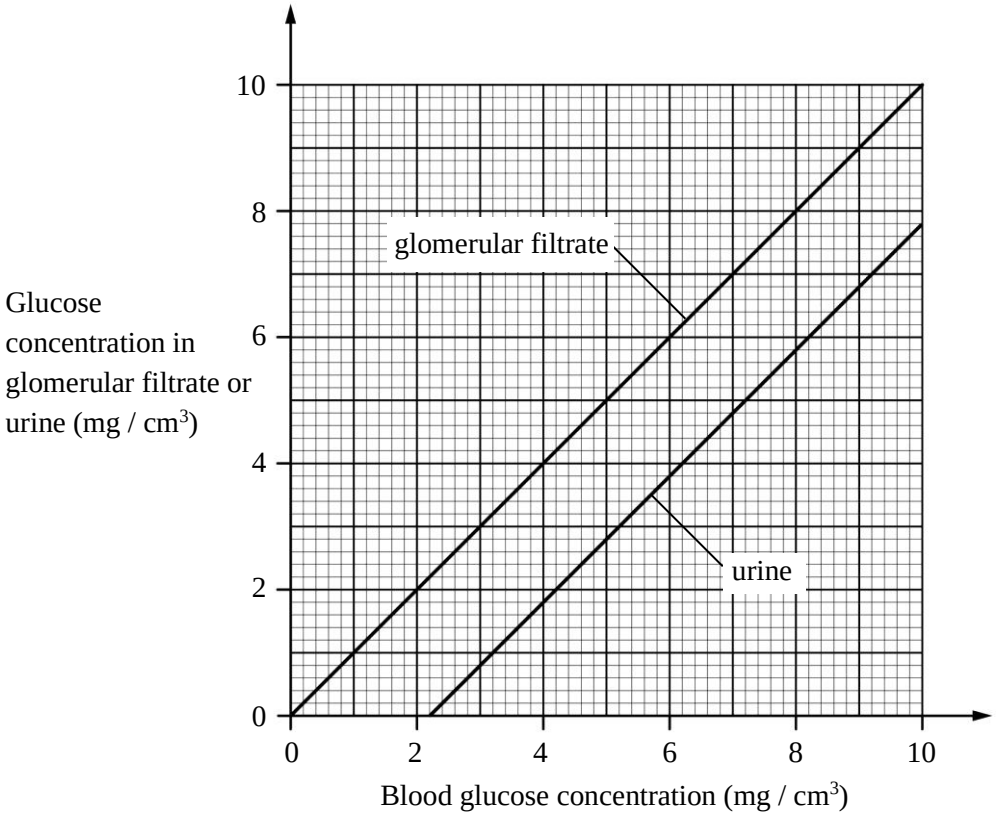
##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Analysing data | 2 |
| (b) | Understanding basic concepts | 2 |
| (c) | Applying concepts | 4 |
| (d) | Applying concepts | 2 |

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|!|BCELB0330000011|!| ★ **Cross topic**

When the blood glucose concentration exceeds a certain threshold value, the kidney will be unable to reabsorb a portion of the glucose in the glomerular filtrate back into the bloodstream. Through an investigation, the threshold value for glucose reabsorption in the kidneys of a mammal was determined. The results are shown in graph below.




(a) Based on the mechanism for the reabsorption of glucose, explain why there is an upper limit to glucose reabsorption in the kidneys. (3 marks)

(b) With reference to the graph above, state and explain the changes in the glucose concentration in urine as the blood glucose concentration increases from 0 to 4 mg / $cm^3$. (4 marks)

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(c) Suggest ***two*** causes for the presence of glucose in urine. (2 marks)

##

(a) The reabsorption of glucose relies on active transport. (1)
Active transport requires carrier proteins on the cell membrane. (1)
The number of carrier proteins is fixed. (1)

(b) When the blood glucose concentration increases from 0 to 2.2 mg / $cm^3$, no glucose will be found in the urine (1)
because all glucose will be reabsorbed from the glomerular filtrate. (1)
When the blood glucose concentration increases from 2.2 to 4 mg / $cm^3$, the glucose concentration in urine will also increase. (1)
This is because the rate of glucose reabsorption is now lower than the rate at which glucose enters the glomerular filtrate. (1)

(c) Any ***two*** of the following: (2)
- insufficient insulin secretion
- excessive intake of sugary food
- damage of the Bowman's capsule

(Or other reasonable answers)

(9 marks)

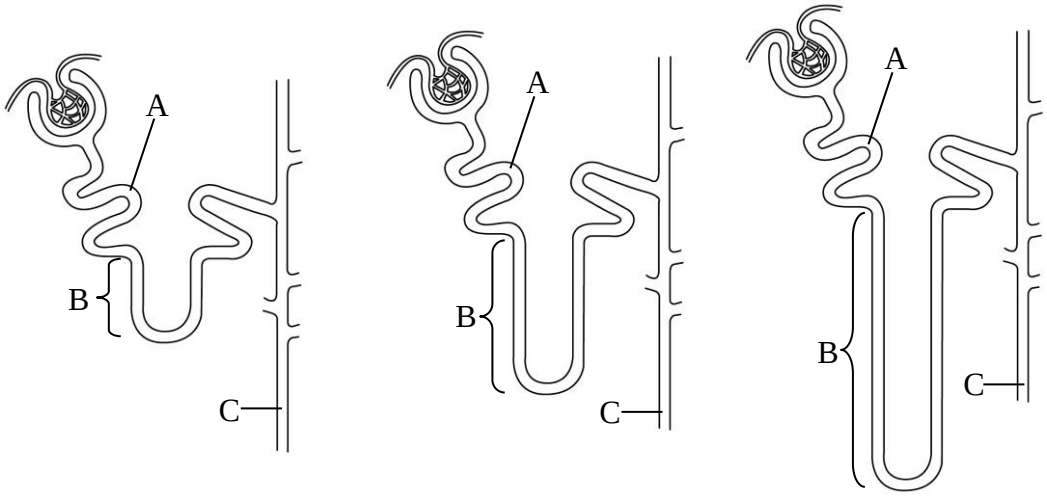
##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Applying concepts | 3 |
| (b) | Applying concepts | 4 |
| (c) | Understanding basic concepts | 2 |

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|!|BCELA0330000012|!| ★★

The diagrams below show the nephrons from the kidneys of three mammals: a river otter, a house cat and a desert fox. (The diagrams are ***not*** drawn to the same scale.)




**nephron of river otter** | **nephron of house cat** | **nephron of desert fox**

(a) Explain how the features of the epithelial cells in structure A affect the composition of the filtrate flowing through structure A. (6 marks)

(b) Name structures B and C. (2 marks)

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(c) (i) Compare the water potential of the urine produced by the three mammals. (2 marks)

(ii) Explain the trend stated in your answer to (i). (2 marks)

(iii) State and explain the relationship between the three mammals' habitats and the length of structure B in the mammals' nephrons. (2 marks)

##

(a) The numerous mitochondria in the epithelial cells of structure A (the priximal convoluted tubule) provide energy (1)
for the active transport of salts / amino acids / glucose into the bloodstream. (1)
This raises the water potential of the filtrate and water is thus returned to the blood through osmosis. (1)
Microvilli on the surface of epithelial cells provide a large surface area (1)
for the reabsorption of urea / glucose / amino acids / salts (any two) by diffusion. (1)
Thus these features reduce the amount of salts, glucose, amino acids, urea and water present in the filtrate. (1)

(b) B: loop of Henle (1)
C: collecting duct (1)

(c) (i) The water potential of the urine produced by the three mammals in descending order:
river otter > house cat > desert fox (2)

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(ii) The longer the loop of Henle is, the greater the proportion of water that will be reabsorbed from the filtrate through osmosis. (1)
This leads to the production of urine with a lower water potential. (1)
(Or vice versa)

(iii) Desert mammals living in dry habitats have longer loop of Henle in order to conserve more water. (1)
Aquatic mammals living in freshwater rivers have shorter loop of Henle in order to reduce water reabsorption and excrete excess water. (1)

(14 marks)

##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Applying concepts | 6 |
| (b) | Analysing data | 2 |
| (c)(i) | Applying concepts | 2 |
| (c)(ii) | Applying concepts | 2 |
| (c)(iii) | Applying concepts | 2 |

|!|BCELB0330000013|!| ★

The first stage in the formation of urine is ultrafiltration, which produces the glomerular filtrate in the Bowman's capsule.

(a) State ***two*** substances that are present in the glomerular filtrate. (2 marks)

(b) The presence of protein molecules in the urine of a person is a sign of kidney disease.

(i) Explain why it is abnormal for protein molecules to appear in urine. (2 marks)

(ii) Explain why the presence of proteins in urine is often a symptom of chronic hypertension (high blood pressure). (2 marks)

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(c) Coffee contains the drug called caffeine, which inhibits the release of ADH. Explain how drinking coffee will affect the volume and solute concentration of urine. (4 marks)

##

(a) Any ***two*** of the following: (2)

- water
- glucose
- amino acids
- mineral salts
- urea

(b) (i) Protein molecules are too large (1)
to pass through the walls of the glomerulus and the Bowman's capsule. (1)

(ii) Hypertension raises the hydrostatic pressure in the glomerulus. (1)
This forces proteins to pass through the capillary walls and the walls of the Bowman's capsule and enter the glomerular filtrate. (1)

(c) The amount of ADH released by the pituitary gland will be reduced, thus causing the walls of the collecting ducts to become less permeable to water. (1)
A smaller proportion of water will be reabsorbed from the glomerular filtrate into the blood. (1)
The volume of urine will increase and (1)
the solute concentration of urine will decrease. (1)

(10 marks)

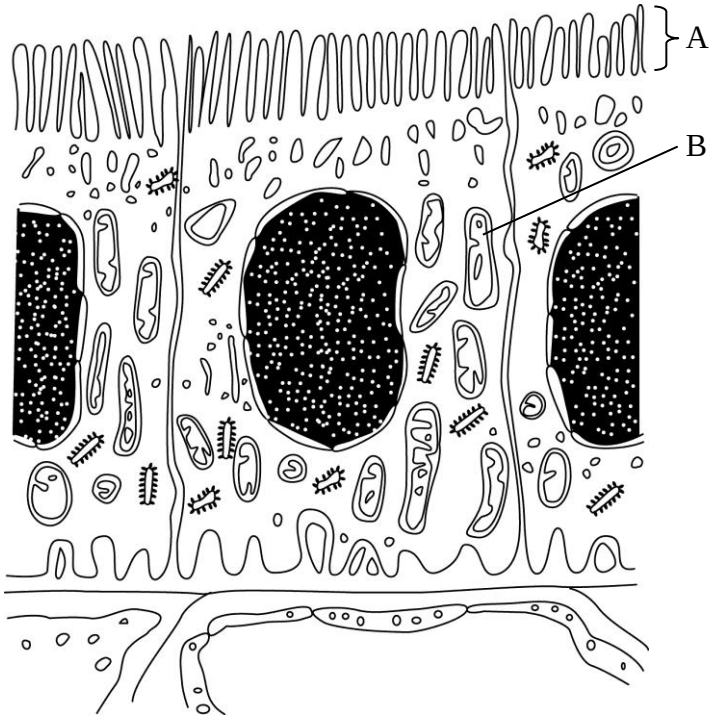
##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Understanding basic concepts | 2 |
| (b)(i) | Understanding basic concepts | 2 |
| (b)(ii) | Applying concepts | 2 |
| (c) | Applying concepts | 4 |

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|!|BCELC0330000014|!|

The diagram below is a partial longitudinal section that shows the cellular structure of the proximal convoluted tubule in a nephron.




(a) Name structures A and B. (2 marks)

(b) State and explain ***two*** adaptations in the cells shown above that facilitate reabsorption. (4 marks)

(c) Outline how water is reabsorbed by the epithelial cells of the proximal convoluted tubule. (3 marks)

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##

(a) A: microvillus (1)

B: mitochondrion (1)

(b) The presence of numerous microvilli (1)

increases the surface area for reabsorption of substances. (1)

Numerous mitochondria provide a rich supply of ATP, (1)

which drives the active transport of certain compounds (e.g. glucose, amino acids and salts). (1)

(c) As salts / glucose / amino acids are absorbed into the epithelial cells by diffusion and active transport, (1)

the cells' water potential decreases compared to the glomerular filtrate /

the cells' solute concentration increases compared to the glomerular filtrate. (1)

Water then enters the epithelial cells by osmosis. (1)

(9 marks)

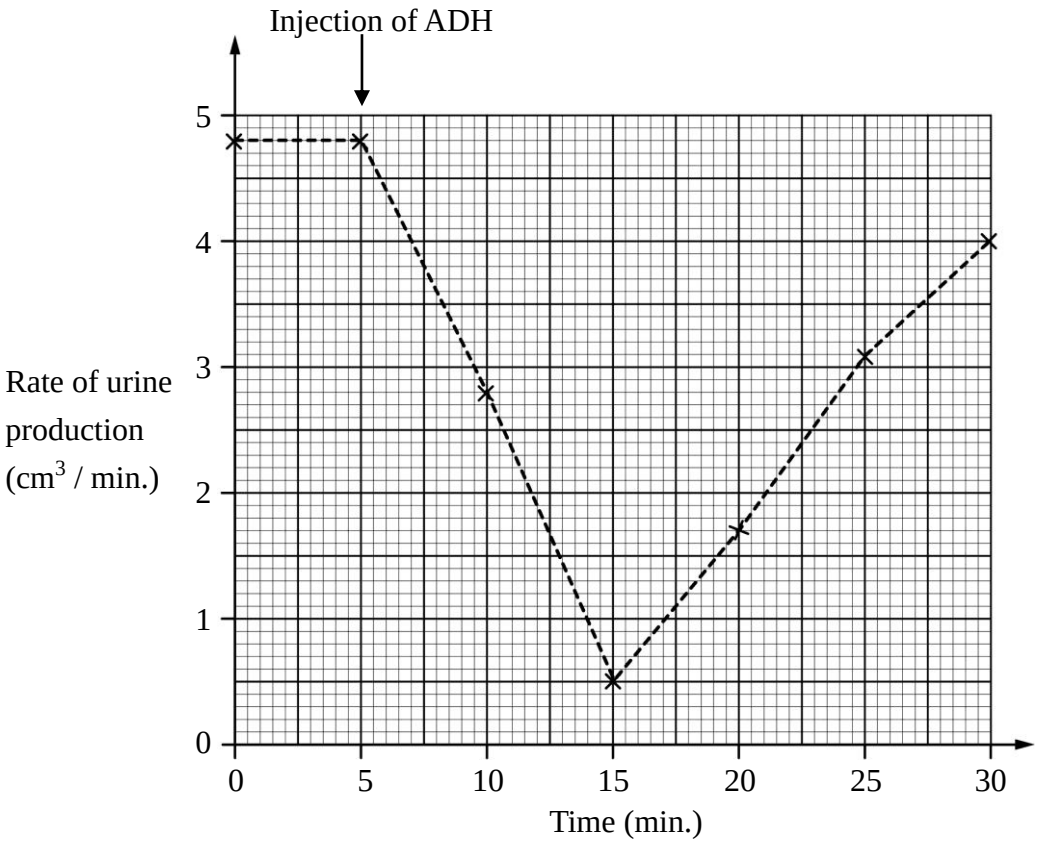
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| | Skills assessed | Marks |
|---|---|---|
| (a) | Analysing data | 2 |
| (b) | Analysing data | 2 |
| | Understanding basic concepts | 2 |
| (c) | Applying concepts | 3 |

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|!|BCELB0330000015|!| ★

Andrew carried out an investigation to determine how antidiuretic hormone (ADH) affects the production of urine in a mammal. He measured the rate of urine production over a period of 30 minutes. Five minutes after measurements began, Andrew injected ADH into a vein of the mammal. The results are shown in the graph below.




(a) Name the body structure that produces ADH. (1 mark)

_____

(b) Referring to the graph above, explain the changes in the rate of urine production after the injection of ADH. (5 marks)

_____

_____

_____

_____

_____

(c) In a similar investigation, Andrew injected a salt solution instead of ADH into the mammal. Explain the effect of the injection on the rate of urine production.

(4 marks)

##

(a) Hypothalamus (1)

(b) The injected ADH increased the permeability of the wall of the collecting ducts to water. (1)

A greater proportion of water in the glomerular filtrate was then reabsorbed into the blood. (1)

Thus a smaller volume of concentrated urine was produced and the rate of urine production decreased between 5 and 15 minutes. (1)

As the increase in ADH was temporary, a smaller proportion of water was reabsorbed into the blood over the following 15 minutes. (1)

This caused a larger volume of urine to be produced and the rate of urine production increased between 15 and 30 minutes. (1)

(c) The salt solution decreased the water potential of the blood, (1)

which was detected by osmoreceptors in the hypothalamus. (1)

The hypothalamus stimulated the pituitary gland to release more ADH. (1)

A smaller volume of concentrated urine was thus produced and the rate of urine production decreased. (1)

(10 marks)

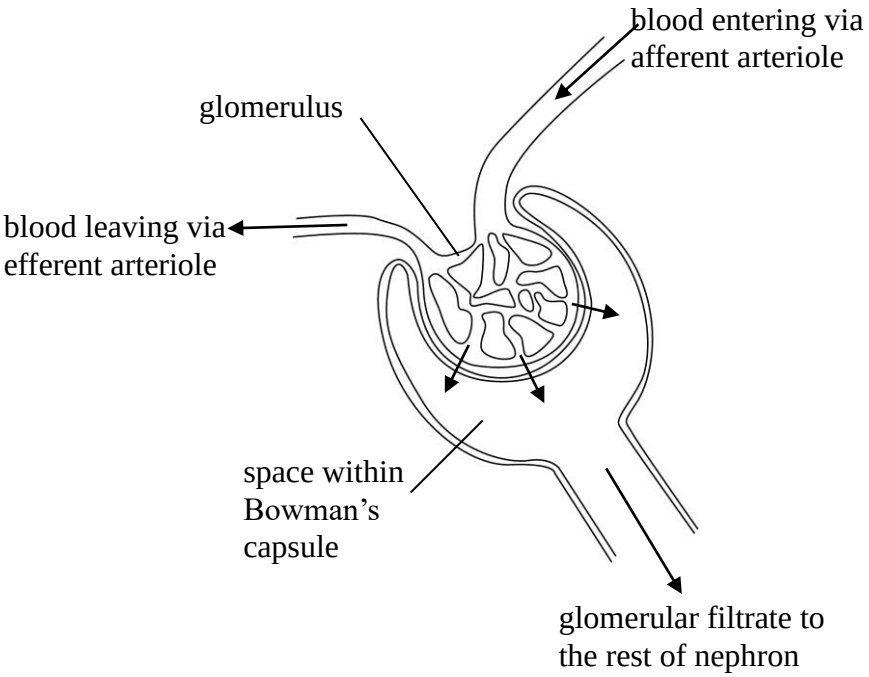
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| | Skills assessed | Marks |
|---|---|---|
| (a) | Understanding basic concepts | 1 |
| (b) | Applying concepts | 5 |
| (c) | Applying concepts | 4 |

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|!|BCELA0330000016|!| ★★

The diagram below shows part of a nephron from a human kidney.




Atrial natriuretic factor (ANF) is a hormone that increases the diameter of the afferent arteriole and decreases the diameter of the efferent arteriole.

(a) Describe and explain how ANF will affect the rate of ultrafiltration. (2 marks)

(b) Vivian measured both the volume and urea concentration of the urine she produced on two different days. The results are shown in the following table.

| **Day** | **Volume of urine produced ($dm^3$)** | **Concentration of urea ($g/dm^3$)** |
|---|---|---|
| 1 | 1.3 | 16 |
| 2 | 1.8 | 9 |

(i) Calculate the percentage change in the urea concentration of the urine between day 1 and day 2. Show your working. (2 marks)

(ii) Suggest ***two*** possible reasons that would account for the lower urea concentration of the urine on day 2. (2 marks)

(c) In the proximal convoluted tubule, large volumes of water are reabsorbed from the glomerular filtrate into the blood. Explain why the concentration of sodium ions remains unchanged but the concentration of urea increases along the proximal convoluted tubule. (4 marks)

##

(a) ANF will increase the rate of ultrafiltration, (1)
because it raises the hydrostatic pressure of the blood in the glomerulus. (1)

(b) (i) Percentage change = (9−16) / 16 × 100% (1)
= −43.75% (1)

(ii) Any ***two*** of the following: (2)

- On day 2, less protein in the diet reduces the amount of excess amino acids.
- On day 2, more water in the diet results in more water in the urine.
- On day 1, dehydration / lower water intake results in less water in the urine.
- On day 2, a greater metabolic demand for amino acids due to training / growth / repair / pregnancy reduces the amount of excess amino acids.
- On day 2, urea loss through sweating reduces the amount of urea in the urine.

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(c) Sodium ions are reabsorbed into the blood by active transport. (1)
This causes the reabsorption of water through osmosis by an amount proportional to that of the reabsorbed sodium ions. (1)
In contrast, a comparatively lower proportion of urea is reabsorbed into the blood (1)
by diffusion. (1)

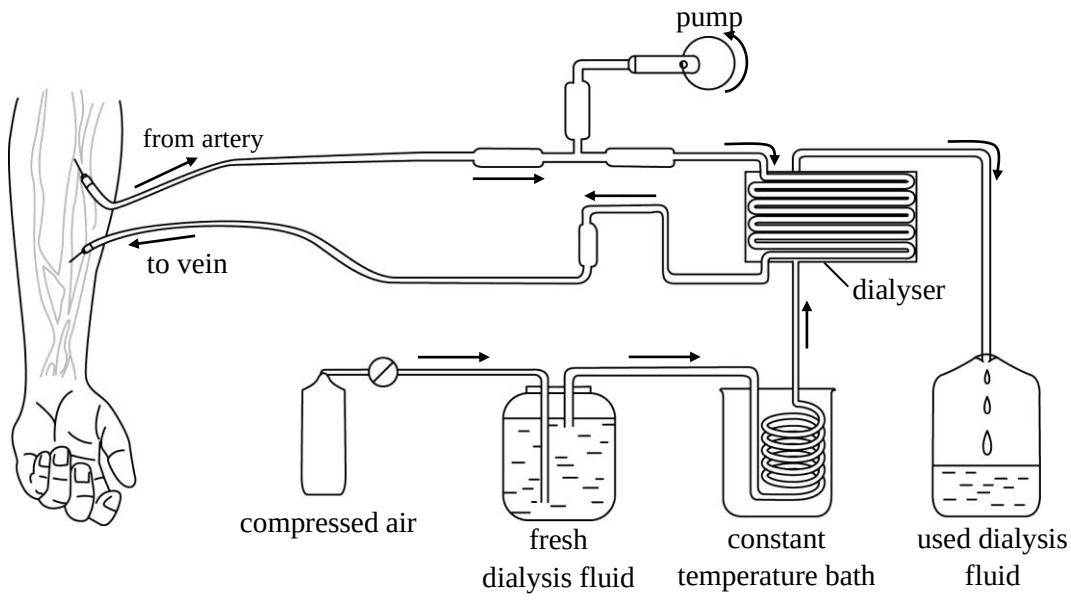
(10 marks)

##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Applying concepts | 2 |
| (b)(i) | Mathematical skills | 2 |
| (b)(ii) | Applying concepts | 2 |
| (c) | Applying concepts | 4 |

|!|BCELB0330000017|!| ★ **STSE**

The diagram below shows a method used to treat kidney failure.




(a) State ***two*** effects that kidney failure has on the body. (2 marks)

(b) The dialyser uses tubing with cellophane membranes.

(i) What is the notable feature of cellophane membranes? (1 mark)

(ii) Explain how the feature in (b)(i) can facilitate the treatment method shown. (3 marks)

(c) Suggest ***two*** disadvantages for the treatment above. (2 marks)

(d) Blood must be properly regulated before it is transfused back into the patient. Name ***two*** factors that require such regulation and state their physiological importance. (4 marks)

(e) An anticoagulant is added to the blood before the treatment above begins. Why? (1 mark)

##

(a) Metabolic wastes accumulate in the blood. (1)
The water balance in the body is upset. (1)

(b) (i) It is differentially permeable. (1)

(ii) Metabolic waste (e.g. urea) can pass through the pores of the cellophane membrane (1)
and diffuse into the dialysis fluid along the concentration gradient. (1)

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Conversely, larger molecules like red blood cells cannot pass through the membrane and are prevented from leaving the bloodstream. (1)

(c) Any ***two*** of the following: (2)

- risk of bacterial infection
- limited mobility when using the dialyser
- large expense / cost

(Or other reasonable answers)

(d) Any ***two*** sets of the following or other reasonable answers: (4)

| Factor regulated | Physiological Importance |
|---|---|
| Blood temperature | Maintaining the optimum temperature for cell metabolism |
| Blood pH | Maintaining the optimum pH for cell metabolism |
| Bacterial / pathogen level | bacteria / pathogens can cause harm to the body |

(e) The anticoagulant prevents blood clotting and ensures a smooth blood flow during haemodialysis. (1)

(13 marks)

##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Understanding basic concepts | 2 |
| (b)(i) | Understanding basic concepts | 1 |
| (b)(ii) | Understanding basic concepts | 3 |
| (c) | Applying concepts | 2 |
| (d) | Applying concepts | 4 |
| (e) | Applying concepts | 1 |

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